

Preferentially Constrained Information Traversal: ORBIT Foundational Construction

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Gemini 2 and GPT 5 jointly generated this work; the author has since applied edits.

Abstract

This construction introduces ORBIT, a class for preferentially constrained information traversal in machine learning. The construction distinguishes a physical domain of realisations and constraints from a discursive domain of representations, predicates, and transformations. Machine-learning experiments are coupled objects spanning these domains. To reflect the reality of perception and control recursive ambiguity, one restricts observer access to boundary-relative, partial, and potentially distorting observation maps. Hierarchies (e.g. self-reference and access) reside internally within the discursive domain.

The interaction of ML problems, preferences, and constraints then induces graded admissibility over discourse-level objects with feasibility constraints projected from the physical domain. These ingredients define a class of boundary-integrated traversers, including minimum-divergence local traversal as a special case, and motivate conditions for local, component-wise, and global reachability, and their failure modes. ORBIT is therefore proposed as a foundational construction for analysing traversal and constraint integration in coupled physical-discursive learning systems.

1 Preparatory Construction

The ORBIT class is a class of ‘ontologically recursive boundary-integrated traversers’. This note describes how to construct these traversers in a minimal way¹, separating ontological postulates, observer-specific access conditions, graded admissibility objects, and traversal targets. Later sections may prove reachability and failure results under explicit additional assumptions.

Construction 1 (ORBIT preparatory foundational claim). *Construct a class of ontologically recursive boundary-integrated traversers by the following staged specification.*

(1) **Ontological separation.** *Postulate two analytically distinct but coupled domains:*

$$\mathcal{P} \text{ (physical domain),} \quad \mathcal{D} \text{ (discursive domain).} \quad (1)$$

Here, $\mathcal{P}$ contains ML experiment realisations, material resources and their dynamics, and physical constraints. $\mathcal{D}$ contains representations, problem statements, predicates, operators, admissibility objects, and discourse-level transformations. One requires no stronger metaphysical claim than analytic distinctness and coupling for subsequent results.

¹Not yet fully axiomatised.

(2) **Cross-domain status of machine-learning problems.** *This construction does not assume machine-learning problems arise exclusively in either $\mathcal{P}$ or $\mathcal{D}$. The construction instead treats an ML problem as a coupled construct. The problem's formal specification lies in $\mathcal{D}$, its implementation occurs in $\mathcal{P}$, and its evidential accessibility depends on an observer.*

(3) **Machine-learning problem as a coupled object.** *Define a machine-learning problem as a coupled object*

$$M = (\Pi, R, I, C; \kappa_{\text{prob}}), \quad (2)$$

where $\Pi \in D$ is the problem specification, $R \in \mathcal{R} \subseteq D$ is a representational regime, $I \subseteq D \times P$ is an implementation relation, C is a constraint family spanning D and/or P , and κ_{prob} records the minimal invariant signature required for problem-identity under admissible variation. An experiment is an enacted instance of M under observer-relative access.

(3a) **Minimal identity invariants of coupled objects.** *Let*

$$Z \subseteq D \times P \quad (3)$$

denote the class of coupled physical-discursive objects accessible to the framework. For each object type

$$\tau \in \{\text{prob}, \text{exp}, \text{sol}\}, \quad (4)$$

let

$$\kappa_{\tau} : Z \rightarrow \mathcal{I}_{\tau} \quad (5)$$

be an identity-invariant map into a corresponding invariant space $\mathcal{I}_{\tau}$. The value $\kappa_{\tau}(z)$ encodes the minimal coupled structure whose preservation is required for z to remain the same machine-learning problem, experiment-type, or solution-type, respectively. A variation

$$z_t \rightsquigarrow z_{t+1} \quad (6)$$

is type-preserving for τ only if

$$\kappa_{\tau}(z_{t+1}) = \kappa_{\tau}(z_t), \quad (7)$$

or, where exact preservation is too strong, if the corresponding invariant discrepancy remains below an admissible tolerance. ORBIT therefore permits variation only over coupled degrees of freedom not fixed by the active identity invariants and boundary constraints.

- (4) **Observer-relative access and anti-regress boundary.** Let $\mathcal{O}$ be a set of observers. For each $o \in \mathcal{O}$, define an evidence state E_o and observation maps

$$\Omega_o^P : \mathcal{P} \rightarrow E_o, \quad \Omega_o^D : \mathcal{D} \rightarrow E_o. \quad (8)$$

These maps are generally partial, lossy, delayed, and structure-distorting. Define an observer boundary

$$B_o \subseteq \mathcal{P} \cup \mathcal{D} \quad (9)$$

such that observer o interacts directly only through B_o . Recursive or higher-order structures may exist within $\mathcal{D}$, but no external observer-of-observer regress is required unless it is itself re-encoded as an object in $\mathcal{D}$.

- (5) **Incomplete observation and disturbance.** Observational access is typically incomplete. Unless all relevant realisations and representations are jointly available, the observer accesses only a restricted image of the system. This incompleteness may induce selection effects, distortion, or intervention-sensitive changes in either $\mathcal{P}$ or $\mathcal{D}$. Accordingly, observation is modeled as epistemically restricted and, where appropriate, state-affecting.
- (6) **Self-representation and boundary-accessible hierarchy.** Assume that $\mathcal{D}$ may contain self-representational layers or transverse hierarchies:

$$\{D_\alpha\}_{\alpha \in A}, \quad D_\alpha \subseteq \mathcal{D}. \quad (10)$$

For each observer o , define a boundary-accessible discursive substructure

$$D_o^* \subseteq \mathcal{D} \quad (11)$$

through which direct observer access occurs. Thus, self-reference and representational recursion may exist within $\mathcal{D}$. One arrests regress by restricting direct observer coupling to D_o^* .

- (6a) **Hierarchy classes, flat encodings, and structured accessibility.** Let

$$\mathcal{C} = \{\text{opt}, \text{obs}, \text{rep}, \text{acc}\} \quad (12)$$

denote the active hierarchy classes, corresponding respectively to optimisation, observer, representational, and access hierarchies. For each $c \in \mathcal{C}$, let

$$H_c = (X_c, \triangleleft_c, \lambda_c) \quad (13)$$

be a hierarchy instance, where X_c is a collection of objects, $\triangleleft_c$ is the class-specific higher-order relation, and

$$\lambda_c : X_c \rightarrow \mathbb{N} \quad (14)$$

is a level map wherever such indexing is admissible. Require that each hierarchy instance admits both:

(i) a flat discourse-level encoding

$$F_c : H_c \rightarrow D, \quad (15)$$

through which one treats the hierarchy as a single selectable object in discourse; and

(ii) structured access operators

$$A_{c,k} : D \dashrightarrow 2^{X_c}, \quad (16)$$

where $A_{c,k}(F_c(H_c))$ retrieves the level- k objects of H_c whenever such access is available at the active observer boundary. Optionally admit level-wise update operators

$$U_{c,k} : A_{c,k}(F_c(H_c)) \dashrightarrow A_{c,k}(F_c(H_c)). \quad (17)$$

An ORBIT traversal may therefore: (a) select a hierarchy instance H_c as a focal object relative to the active problem, (b) traverse its flat encoding $F_c(H_c)$ as an object in D , and (c) traverse within and across its levels through the structured access and update operators.

A representational regime R_t is said to be hierarchy-sufficient for an active hierarchy instance H_c if it supports both the flat encoding $F_c(H_c)$ and the structured operators required for all currently relevant levels. If the active regime is not hierarchy-sufficient, admissible representation change may be required.

(7) Fixed and variable representational regimes. Let R_t denote the active representational regime at update step t . Distinguish

$$R_t = R \quad \forall t \quad (\text{fixed-representation optimisation}) \quad (18)$$

from

$$R_t \neq R_{t'} \quad \text{for some } t \neq t' \quad (\text{representation-changing optimisation}). \quad (19)$$

Representation change is admissible in particular when the active regime fails to be hierarchy-sufficient for the focal hierarchy instance under Item (6a). Thus, representational variation is one special mechanism by which ORBIT may recover selectable flat encodings, structured level-access, or level-wise update capacity not available in the current regime.

(8) **Graded solution structure.** For each problem specification Π , let

$$\preceq_{\Pi} \subseteq D \times D \quad (20)$$

be a preorder on discourse-level objects, where

$$x \preceq_{\Pi} y \quad (21)$$

means that y is at least as admissible as x relative to Π . Learning then seeks, exactly or approximately, objects that are maximal or near-maximal under $\preceq_{\Pi}$. The mechanism inducing $\preceq_{\Pi}$ may be specified separately, and may depend on admissibility objects, predicates, projected feasibility constraints, or other discourse-level structure. Hence, discourse-level objects are not merely solutions/non-solutions in a binary sense; each $x \in \mathcal{D}$ may possess a degree of admissibility $\sigma_{\Pi}(x)$ relative to Π .

(9) **Physical feasibility as projected constraint.** Physical feasibility as projected constraint. Let C_P denote the active family of physical constraints in P . Through the implementation relation and the active observer boundary, these induce a projected feasibility structure on discourse-level objects. At minimum, this may be given by a feasibility predicate

$$\text{Feas}_P : D \rightarrow \{0, 1\}, \quad (22)$$

where $\text{Feas}_P(x) = 1$ means that x admits an acceptable physical instantiation. More generally, physical feasibility may induce a preorder

$$\preceq_P^{\phi} \subseteq D \times D, \quad (23)$$

where

$$x \preceq_P^{\phi} y \quad (24)$$

means that y is at least as physically feasible as x . Thus, physical feasibility need not be represented numerically; it may instead act as a projected constraint or comparison structure on D . The overall admissibility preorder $\preceq_{\Pi}$ may then depend in part on this projected physical-feasibility structure together with further discourse-level constraints.

(10) **Structure-preserving cross-representation maps.** Where representational regimes preserve designated invariants, admit structure-preserving maps²

²A categorical formulation may be useful here, for example by treating representational regimes as categories and cross-representation translations as functors preserving designated structure. However, no categorical commitment is required at this stage. The present construction admits weaker, pre-categorical notions of structure-preserving correspondence wherever these suffice.

$$f : X \rightarrow Y, \quad X, Y \subseteq \mathcal{D}, \quad (25)$$

which preserve relevant symmetries, relations, or admissibility structure. In special cases such maps may define valid dual access between representations, thereby permitting objects in one regime to correspond to operators, predicates, or transformations in another.

(11) **Operational semantics as a discourse-level special case.** Operational semantics supplies a canonical instance in which symbolic or linguistic objects acquire meaning via admissible transformation or execution rules. Thus, language provides one special case of discourse-level objects whose operational behavior supports object–predicate or object–operation identifications.

(12) **Discursive closure and self-reference.** Under suitable self-referential and update conditions, the discursive component of an experiment may exhibit operational closure: objects in $\mathcal{D}$ may generate, evaluate, and transform further objects in $\mathcal{D}$ using internally available structure. This closure is internal to $\mathcal{D}$ and is not to be conflated with the observer’s limited hierarchical access.

(13) **Constraint compatibility and boundary-integrated admissibility.** Physical feasibility is not the only active constraint. Let

$$C_{\Pi,o} = \{c_1, \dots, c_m\}$$

denote the active family of primal constraint objects relative to problem specification Π and observer o . Assume that, wherever boundary-accessible, each c_i admits an operationalisation

$$\delta_{o,i}(c_i),$$

through which it can be applied to discourse-level candidates. Each such operationalisation induces a preorder

$$\preceq_{i,\Pi,o} \subseteq D \times D,$$

where

$$x \preceq_{i,\Pi,o} y$$

means that y is at least as admissible as x relative to the i -th active constraint. If these induced preorders are jointly comparable, or if a comparison rule over them has been specified, define an aggregate admissibility preorder

$$\preceq_{\Pi,o}^* \subseteq D \times D.$$

In the minimal Pareto case,

$$x \preceq_{\Pi,o}^* y \iff x \preceq_{i,\Pi,o} y \text{ for all } i = 1, \dots, m.$$

Boundary-induced admissibility therefore refers to admissibility comparison induced by operationally accessible constraints, including constraints projected through observer and implementation boundaries.

(14) **General traversal rule.** For a fixed observer $o \in O$, let

$$\preceq_{\Pi,o}^*$$

be the active aggregate admissibility preorder, and let

$$\mathcal{R}_o : D \rightarrow 2^D$$

be an admissibly reachable-candidate correspondence, where $\mathcal{R}_o(x)$ is the class of candidates available from x under the active boundary, representational, and implementation constraints. Define a traversal rule

$$x_{t+1} \in \mathcal{T}_o(x_t; \preceq_{\Pi,o}^*, \mathcal{R}_o, B_o),$$

where $\mathcal{T}_o$ selects an admissible successor from $\mathcal{R}_o(x_i)$. No locality requirement is imposed at this level.

(14a) **Local traversal as a special case.** If a local rule is desired, let

$$\mathcal{L}_o : D \rightarrow 2^D$$

be a local-candidate correspondence with

$$\mathcal{L}_o(x) \subseteq \mathcal{R}_o(x)$$

for each $x \in D$. Locality need not be metric; it may instead be semantic, syntactic, or representation-relative, provided it determines a restricted candidate class near the focal object. If, in addition, one specifies a perturbation preorder

$$\trianglelefteq_x \subseteq \mathcal{L}_o(x) \times \mathcal{L}_o(x),$$

where

$$y \trianglelefteq_x z$$

means that y is no greater a perturbation of x than z , then a minimum-divergence local traversal rule may be defined by selecting $\trianglelefteq_x$ -minimal admissibility-improving or admissibility-preserving successors within $\mathcal{L}_o(x)$.

(15) **Reachability under persistence and compatibility assumptions.** Suppose admissibility objects persist sufficiently across updates, representation transitions remain compatible, and admissible regions are appropriately connected. Then traversal may achieve forms of reachability, including local reachability, component reachability, and, under stronger assumptions, global reachability across objects and representations. These conditions are proposition targets rather than automatic consequences of the framework.

(16) **Failure modes of global reachability.** Global reachability may fail through disconnected admissible components, boundary opacity, nonpersistent feasibility, incompatible representation changes, symmetry-breaking transitions, local minima, divergence traps, observer-induced distortion, or dynamic drift in the problem specification itself. Hence, global reachability is conditional and non-generic.

(17) **Definition of ORBIT.** Define ORBIT to be the class of traversers $\mathfrak{T}$ satisfying the preceding construction: they act over coupled physical and discursive domains, permit discourse-level self-reference, integrate observer- and implementation-boundary constraints into admissibility, and traverse by admissibility-structured update rules. Formally, an ORBITAL system is any tuple

$$\mathfrak{O} = (\mathcal{P}, \mathcal{D}, \mathcal{O}, \{B_o\}_{o \in \mathcal{O}}, M, \sigma, \phi, \{a_i\}_{i=1}^m, \mathcal{A}, \mathfrak{T}) \quad (26)$$

satisfying Items (1)–(16).

Remark 1 (Interpretive status). *Construction 1 is a staged framework claim. Items (1), (4), and (5) function primarily as ontological or epistemic postulates; Items (3), (8), (9), (13), and (17) supply core definitions; Items (6), (7), (10), (11), and (12) specify structural enrichments; Items (14)–(16) identify traversal and reachability targets to be proved under additional assumptions.*

2 ORBIT: Operational Completion

The foundational construction specifies the ontological and admissibility structure of ORBIT. One must now add the minimal operational elements required to recover representation switching, constraint integration, and comparative performance claims.

Definition 1 (Invariant object and representation library). *Let $\mathcal{I}$ denote an invariant object space, whose elements encode the representation-independent content of a machine-learning experiment up to the chosen equivalence notion. Let*

$$\mathcal{R} = \{r_1, \dots, r_K\} \quad (27)$$

be a candidate representation library, taken as elements of the discursive domain. For each $r \in \mathcal{R}$, let $X_r \subseteq \mathcal{D}$ denote the state space of discourse objects expressed in representation r , and let

$$\psi_r : X_r \rightarrow \mathcal{I} \quad (28)$$

be an abstraction map into invariant space.

Definition 2 (Representation-conditioned fixed-budget value). *For a budget $B > 0$ and a representation $r \in \mathcal{R}$, let*

$$J(r; B) \quad (29)$$

denote the best attainable value of the chosen objective when traversal is restricted to X_r for the entire budget horizon. The representation advantage at budget B is

$$\Delta_{\mathcal{R}}(B) := \max_{r \in \mathcal{R}} J(r; B) - \min_{r \in \mathcal{R}} J(r; B). \quad (30)$$

Definition 3 (Constraint integration). *For a fixed observer $o \in \mathcal{O}$, let $\mathfrak{C}_t = \{c_{1,t}, \dots, c_{m,t}\}$ be the active constraint family at time t . For each $r \in \mathcal{R}$ and each active constraint $c_{i,t}$, define a representation-specific integration map*

$$\iota_{r,i,t} : X_r \rightarrow [0, \infty] \quad (31)$$

whose value measures the degree of violation induced by $c_{i,t}$ in representation r . Hard violations may be encoded by $+\infty$. Given nonnegative preference weights $\lambda_{i,t}$, define the representation-specific aggregate violation potential

$$V_r(x; t) = \sum_{i=1}^m \lambda_{i,t} \iota_{r,i,t}(x), \quad x \in X_r. \quad (32)$$

The corresponding active boundary is

$$B_{r,t}^{(o)} = \{x \in X_r : V_r(x; t) < \infty\}, \quad (33)$$

or, more generally, the admissible sublevel region induced by the active hard and soft constraints.

Definition 4 (Internal update and semantic bridge). *For each $r \in \mathcal{R}$, let*

$$U_r : X_r \rightarrow X_r \quad (34)$$

*denote an admissible internal update rule. For each ordered pair $(r, r') \in \mathcal{R} \times \mathcal{R}$, a **semantic bridge** is a partial map*

$$B_{r \rightarrow r'} : \text{Dom}(B_{r \rightarrow r'}) \subseteq X_r \rightarrow X_{r'}. \quad (35)$$

*The bridge is said to be ε -**valid at** $x \in X_r$ if*

$$d_{\mathcal{I}}(\psi_{r'}(B_{r \rightarrow r'}(x)), \psi_r(x)) \leq \varepsilon, \quad (36)$$

for a chosen invariant discrepancy $d_{\mathcal{I}}$. Thus, internal updates move within a representation, while bridge maps transport a state between representations with controlled invariant distortion.

Definition 5 (Representation graph and local reachability). *Define the directed representation graph*

$$G_{\mathcal{R}} = (\mathcal{R}, E), \quad (r, r') \in E \iff B_{r \rightarrow r'} \text{ is admissible on a nontrivial domain.} \quad (37)$$

*One says that $\mathcal{R}$ is **locally bridge-reachable** if every $r \in \mathcal{R}$ has at least one outgoing or incoming admissible bridge edge. One says that $\mathcal{R}$ is **globally bridge-reachable** if $G_{\mathcal{R}}$ is strongly connected.*

Definition 6 (Local diagnostics and switching policy). *Let*

$$\delta_t(r, x) \quad (38)$$

denote the collection of locally available diagnostics at time t for state $x \in X_r$, for example current violation potential, local progress estimates, bridge distortion estimates, and switching overhead estimates. An ORBIT policy is a measurable decision rule

$$\pi_t : \delta_t(r_t, x_t) \mapsto a_t \quad (39)$$

whose action a_t is either an internal update in the current representation or a switch to a new representation through an admissible bridge. The induced state evolution is

$$(r_{t+1}, x_{t+1}) = \begin{cases} (r_t, U_{r_t}(x_t)), & a_t = \text{stay}, \\ (r', B_{r_t \rightarrow r'}(x_t)), & a_t = \text{switch to } r'. \end{cases} \quad (40)$$

An ORBIT traversal is therefore a coupled sequence

$$\gamma = \{(r_t, x_t)\}_{t \geq 0} \quad (41)$$

through internal states and representational regimes.

Definition 7 (Dynamic value and regret). Let $J(\text{ORBIT}; B)$ denote the attained objective value of an ORBIT policy under budget B . Let

$$r^*(B) \in \arg \max_{r \in \mathcal{R}} J(r; B). \quad (42)$$

The regret of ORBIT relative to the best fixed representation in hindsight is

$$\text{Regret}_{\mathcal{R}}(B) = J(r^*(B); B) - J(\text{ORBIT}; B). \quad (43)$$

Assumption 1 (Diagnostic sufficiency). Assume that the local diagnostics are sufficiently informative to distinguish, with bounded error, between (i) internal updates whose expected progress has decayed below tolerance and (ii) admissible switches whose expected gain exceeds switching cost and bridge distortion.

Proposition 1 (Representation-advantage target). Suppose $\Delta_{\mathcal{R}}(B) > 0$ and Assumption 1 holds. Then there exists an ORBIT policy class for which

$$J(\text{ORBIT}; B) \geq \max_{r \in \mathcal{R}} J(r; B) - \varepsilon_B, \quad (44)$$

where ε_B depends on diagnostic error, switching overhead, and bridge distortion. In particular, whenever the representation advantage is appreciable and these errors are controlled, dynamic traversal can outperform a representation chosen a priori.

Proposition 2 (Everywhere-cursed limit). If $\Delta_{\mathcal{R}}(B) \approx 0$, then representation changes do not unlock a material fixed-budget advantage. In this regime, any well-behaved ORBIT policy should satisfy

$$\text{Regret}_{\mathcal{R}}(B) = O(\text{switching overhead} + \text{diagnostic error}), \quad (45)$$

and should tend toward policy stabilization rather than persistent switching.

Proposition 3 (Bridge-mediated reachability target). Assume:

1. each admissible component of each X_r is internally reachable under U_r ;

2. $G_{\mathcal{R}}$ is strongly connected;

3. bridge distortion remains below the admissibility tolerance required for continuation of traversal.

Then any admissible object lying in a bridge-connected admissible component of the total state space

$$X_{\mathcal{R}} := \bigsqcup_{r \in \mathcal{R}} X_r \tag{46}$$

is reachable by a finite alternating sequence of internal updates and admissible bridge moves. Failure of any one of these conditions yields a corresponding reachability obstruction.

Remark 2 (Interpretation). *Section 2 extends the foundational construction by making explicit the representation library, representation-conditioned boundaries, semantic bridges, local switching policy, and best-fixed comparison class. The earlier construction defines what ORBIT is; the present section defines how an ORBIT system moves and how its benefit is to be judged.*